

A. HAYDEÉ CONTRERAS PERUYERO

SNII Nivel I | Postdoctoral fellowship at Center of Sciences Mathematics - UNAM | Topological and Geometric Data Analysis | Bioinformatics | Geometric Group Theory | Global Riemannian Geometry | Quantum Geometry and Topology

EDUCATION

- 2021
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2016

Ph. D. in Science in Mathematics
Instituto de Matemáticas, UNAM  Mexico City, Mexico

- Dr. Pablo Suárez Serrato
 - Thesis: Rigidez Cuasi-Isométrica en Variedades de Gráficas Superiores
- 2021
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2020

Specialty in Applied Statistics
Instituto de Investigaciones en Matemáticas Aplicadas y en Sistemas  Mexico City, Mexico

- Grade: 9.25/10.
- 2016
|
2014

Master of Science in Mathematics
Instituto de Matemáticas, UNAM  Mexico City, Mexico

- Dr. Pablo Suárez Serrato
 - Thesis: Sobre la geometría y la topología de las variedades de gráficas
 - Grade: 9.67/10.
- 2012
|
2007

Bachelor's Degree in Mathematics
Universidad Veracruzana  Veracruz, Mexico

- Dr. Jorge Luis López López
 - Thesis: Triangulaciones de 3-variedades hiperbólicas que fibran sobre el círculo con fibra el toro con un agujero
 - Grade: 9.4/10.

APPOINTMENTS

- current
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2025

Postdoctoral research SECIHTI 2025
Centro de Ciencias Matemáticas, UNAM  Michoacan, Mexico

- Dr. Jesús Hernández Hernández
 - Análisis topológico de datos aplicado a datos genómicos y Criptografía con teoría geométrica de grupos

CONTACT

-  haydeeperuyero@gmail.com
-  <https://haydeeperuyero.github.io/>
-  [HaydeePeruyero](#)
-  [Google Scholar Profile](#)

PROGRAMMING LANGUAGES

Expertise: R and Rstudio, Python, Git/Github, Bash
Familiarity: SQL and PowerBI

LANGUAGES

Spanish - Native
English - B2
Korean - A2

ABOUT ME

I work at the intersection of geometry, topology, and bioinformatics, applying TDA and ML to explore patterns in biological data. I'm also passionate about teaching and programming.

OTHER CV VERSIONS

The short version of my CV can be found in this link [Short CV](#)

The Spanish simplified version of my CV is available at this link [Spanish CV](#)

Last updated on 2025-11-12.

- 2025
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2024

Postdoctoral research DGAPA UNAM 2024
 Centro de Ciencias Matemáticas, UNAM

Michoacan, Mexico

- Dr. Daniel Juan Pineda, Dra. Nelly Sélem Mojica, Dr. Jesús Hernández Hernández
 - Aplicaciones de análisis topológico de datos e invariantes de cuasi-isometrías
- 2024

2024 Summer Collaborators IAS
 Institute of Advances Studies

Princeton, USA

- Catherine Searle, Megan Kerr, Regina Rotman, Erin Griffin, Isabel Beach
 - Existence Criteria for Compact Homogeneous Einstein Metrics
- 2024
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2023

Postdoctoral research DGAPA UNAM 2023
 Centro de Ciencias Matemáticas, UNAM

Michoacan, Mexico

- Dr. Daniel Juan Pineda, Dra. Nelly Sélem Mojica, Dr. Jesús Hernández Hernández
 - TDA y Machine learning aplicado a Metagenómica e invariantes de Variedades de Gráficas
- 2023
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2022

Postdoctoral research CONACyT Ciencia de Frontera 217392
 Centro de Ciencias Matemáticas, UNAM

Michoacan, Mexico

- Dr. Jesús Hernández Hernández
 - Propiedades invariantes de Variedades de Gráficas Superiores
- 2022
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2021

Head of the Department of Basic Sciences
 Instituto Tecnológico de Martínez de la Torre

Veracruz, Mexico



PUBLICATIONS

- 2025

Linear bounds for the lengths of geodesics on manifolds with curvature bounded below

- Beach, I., Contreras-Peruyero, H., Rotman, R., & Searle, C. (2025). The Journal of Geometric Analysis, 35(8), 1-22. <https://doi.org/10.1007/s12220-025-02003-6>
- 2025

Topological data analysis captures Horizontal Gene Transfer in Antimicrobial Resistance gene families among clinically relevant bacteria

- Guerrero-Flores, S., Contreras-Peruyero, H., Ibarra-Rodríguez, J. M., Lovaco-Flores, J. A., Nieto-de la Rosa, F. S., Fontove-Herrera, F., & Sélem-Mojica, N. Front. Microbiol. 16:1461293. <https://doi.org/10.3389/fmicb.2025.1461293>
- 2025

Asymptotic dimension and geometric decompositions in dimensions 3 and 4

- Conteras Peruero, H. and Suárez-Serrato, P. Journal of the Australian Mathematical Society. Published online 2025:1-26. <https://doi.org/10.1017/S1446788725000072>

- 2024 ● **CAMDA 2023: finding patterns in urban microbiomes**
- Contreras-Peruyero, H., Nuñez, I., Vazquez-Rosas-Landa, M., Santana-Quinteros, D., Pashkov, A., Carranza-Barragán, M. E., Perez-Estrada, R., Guerrero-Flores, S., Balanzario, E., Muñoz-Sánchez, V., Nakamura, M., Ramírez-Ramírez, L.L. and Selem-Mojica, N. *Front. Genet.* (2024) 15:1449461. <https://doi.org/10.3389/fgene.2024.1449461>
- 2024 ● **Meeting the challenge of genomic analysis: a collaboratively developed workshop for pangenomics and topological data analysis**
- Contreras-Peruyero, H., Guerrero-Flores, S., Ziri6n-Martínez, C., Mejía-Ponce, P. M., Navarro-Miranda, M., Lovaco-Flores, J.A., Ibarra-Rodríguez, J.M., Pashkov, A., Licona-Cassani, C. and Sélem-Mojica, N. *Bioinformatics Advances*, 2024;:, vbae139 <https://doi.org/10.1093/bioadv/vbae139>
- 2024 ● **Physicochemical Properties of Two Mexican Stingless Bee Honeys to Strengthen Their Biocultural Value**
- Xolalpa-Aroche, A., Hernández-Mena, D. I., Moguel-Chin, W. I., Contreras-Peruyero, H., Rivero-Cruz, B. E., Ortiz-Vázquez, E., & Delgado-Suárez, E. J. (2024). *Revista Brasileira de Farmacognosia*, 1-11. <https://doi.org/10.1007/s43450-024-00566-z>
- 2024 ● **Actinobacteria bioprospecting from ore-forming environments**
- Aguilar, C., Alwali, A., Mair, M., Rodriguez-Orduña, L., Contreras-Peruyero, H., Modi, R., Sélem-Mojica, N., Licona-Cassani, C. & Parkinson, E. I. *Microbial Genomics*, 10(5), 001253. ISSN 2057-5858, <https://doi.org/10.1099/mgen.0.001253>
- 2022 ● **Collapsing and group growth as obstructions to Einstein metrics on some smooth 4-manifolds**
- Contreras Peruyero, H, Suárez-Serrato, P. *New York Journal of Mathematics* 28 (2022), 659–671. <http://nyjm.albany.edu/j/2022/28-27.html>



ACCEPTED PAPERS

- 2025 ● **MicroAgroBiome: A toolkit for exploring specialized metabolism and ecological interactions in rhizosphere microbiomes of cultivated crops**
- Aguilar, C., Fontove-Herrera, F., Pashkov, A., García-Estrada, D.A., Contreras-Peruyero, H., Guerrero-Flores, S., Ramírez-Sánchez, O., Sélem-Mojica, N. To appear at *Nucleic Acids Research*. Accepted September 2025. <https://doi.org/10.1093/nar/gkaf1083>
- 2025 ● **Fractal dimensions of complex networks: advocating for a topological approach**
- Andreeva, R., Contreras-Peruyero, H., Krishnagopal, S., Otter, N., Pascali, M.A. and Thompson, E. Accepted May 2025. To appear at the AWM-IMA Springer volume “Research in Computational Topology”. <https://arxiv.org/abs/2506.15236>



PREPRINTS

2025

• Lengths of the Orthogonal Geodesic Chords on Riemannian Manifolds

• Beach, I., Contreras-Peruyero, H., Griffin, E., Kerr, M., Rotman, R. and Searle, C. <https://arxiv.org/abs/2509.19620>

2025

• Uncharacterized *Acetilactobacillus*-like genera represent the most abundant microorganisms in honeys produced by native stingless bee

• Xolalpa-Aroche, A., Contreras-Peruyero, H., Delgado-Suárez, E.J., Hernández-Mena, D.I., Moguel-Chin, W.I., Rivero-Cruz, J.F., Velarde, R.A., Ortiz-Vázquez, E., Rivero-Cruz, B.E., Lovaco Flores, J.A., Rodríguez Orduña, L., Licona Cassani, C., Barona-Gómez, F., and Sélem-Mojica, N. Submitted June 2025. biorxiv: <https://www.biorxiv.org/content/10.1101/2025.06.25.661387v1>

2022

• On the quasi-isometric rigidity of chambers and walls in cusp-decomposable manifolds

• Contreras Peruyero, H. <https://arxiv.org/abs/2010.03736> Submitted 2022



TEACHING EXPERIENCE

2025

• Multivariate Statistics

ENES Morelia

📍 Michoacan, Mexico

• Undergraduate course, Semester 2026-1

• <https://haydeeperuyero.github.io/MultivariateStatisticalAnalysis/#por-qu%C3%A9-estad%C3%ADstica-multivariada>

2024

• Advanced Statistics Course: Machine Learning and Deep Geometric Learning

Centro de Ciencias Matemáticas, UNAM

📍 Michoacan, Mexico

• Graduate Course

• <https://github.com/CursosPosgradoCCM/ML-GDL>

2024

• Scientific Computing: Scientific computing for data analysis

Centro de Ciencias Matemáticas, UNAM

📍 Michoacan, Mexico

• Graduate Course

• https://haydeeperuyero.github.io/Computo_Cientifico/

2023

• Topology advanced course: Applications of Topological Data Analysis

Centro de Ciencias Matemáticas, UNAM

📍 Michoacan, Mexico

• Graduate Course

• https://github.com/HaydeePeruyero/CursoA_Aplicaciones_TDA

2023	● Statistics seminar: Data Analysis with R Centro de Ciencias Matemáticas, UNAM • Graduate Course • https://haydeeperuero.github.io/Seminario_Estadistica/	📍 Michoacan, Mexico
2023	● Advanced Algebra Course: Group Actions on Trees Centro de Ciencias Matemáticas, UNAM • Graduate Course	📍 Michoacan, Mexico
2022	● Teaching Assistant, Advanced Geometry Course: Hyperbolic Geometry Centro de Ciencias Matemáticas, UNAM • Graduate Course	📍 Michoacan, Mexico
2021	● Teaching Assistant, Analytic Geometry II Facultad de Ciencias, UNAM • Professor José Crispín Ruíz Pantaleón • https://github.com/appliedgeometry/Geometria-Analitica	📍 Mexico City, Mexico
2020	● Teaching Assistant, Analytic Geometry II Facultad de Ciencias, UNAM • Professor José Crispín Ruíz Pantaleón • https://github.com/appliedgeometry/Geometria-Analitica	📍 Mexico City, Mexico
2020	● Teaching Assistant, Differential and Integral Calculus II Facultad de Ciencias, UNAM • Professor Oscar A. Garrido and Arturo Sánchez	📍 Mexico City, Mexico
2020	● Teaching Assistant, Analytic Geometry II Facultad de Ciencias, UNAM • Professor Pablo Suárez Serrato • https://github.com/appliedgeometry/Geometria-Analitica	📍 Mexico City, Mexico
2019	● Teaching Assistant, Analytic Geometry I Facultad de Ciencias, UNAM • Professor Pablo Suárez Serrato • https://github.com/appliedgeometry/Geometria-Analitica	📍 Mexico City, Mexico
2014 2012	● Mathematics and Physics High School teacher Instituto Motolinia	📍 Veracruz, Mexico



SHORT COURSES AND WORKSHOPS

- 2024 • **Web pages with Rmarkdown and Github**
Centro de Ciencias Matematicas (CCM), UNAM 📍 Morelia, Mexico
• Encuentro de R-Ladies Morelia y R-Ladies Querétaro
• <https://r-ladies-morelia.github.io/project/webpage/>
• May 31th - June 1st
- 2024 • **Data analysis - PYTHON, BASH Y GIT**
Centro de Ciencias Matematicas (CCM), UNAM 📍 Morelia, Mexico
• Instructors: Haydeé Peruyero(Git), Shaday Guerrero (Bash), Miguel Magaña(Python)
• <https://miguel-mx.github.io/2024-01-24-ccm-unam-swc/>
• Software Carpentry
- 2024 • **Taller de Metagenómica**
TecNM Campus Chiná 📍 Virtual
• Instructors: Haydeé Peruyero(R and Metagenomics), Shaday Guerrero (Bash), Nelly Sélem(Metagenomics)
• <https://haydeeperuyero.github.io/2024-08-26-metagenomics-workshop-online/>
- 2024 • **Viernes de Bioinformática**
Talk 📍 Queretaro, Mexico
• Creación de Páginas web en R
• November 22th and 29th
- 2023 • **Presentations with Markdown: Xaringan and Quarto reveals**
Centro de Ciencias Matematicas (CCM), UNAM 📍 Virtual event
• R-ladies Morelia
• https://r-ladies-morelia.github.io/Xaringan_RLadies2023/
• May 16th
- 2022 • **dplyr y Rmarkdown con Latex**
Centro de Ciencias Matematicas (CCM), UNAM 📍 Virtual event
• R-ladies Morelia
• May 14th
- 2022 • **R for Reproducible Scientific Analysis**
Centro de Ciencias Matematicas (CCM), UNAM 📍 Virtual event
• Helper for R-ladies Morelia



THESIS SUPERVISION

- current
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2024 • **Master's Thesis Co-advisor Luis Raul Figueroa Martinez**
• Thesis title "Topological Data Analysis for Automated Classification of Multicellular Patterns"
• Posgrado Conjunto en Ciencias Matemáticas de la Universidad Nacional Autónoma de México y la Universidad Michoacana de San Nicolás de Hidalgo

current
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2025

Undergraduate's Thesis Advisor Alejandro Quintero Díaz

• Thesis title: "Aplicaciones de la homología persistente en el estudio de pangenomas y resistomas bacterianos"

• Facultad de Ciencias, Universidad Nacional Autónoma de México

2025
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2024

Master's Thesis Advisor Johanna Atenea Carreón Baltazar

• Thesis title "Analysis of the Distribution of Duplicated Genes in Cyanobacteria"

• Posgrado Conjunto en Ciencias Matemáticas de la Universidad Nacional Autónoma de México y la Universidad Michoacana de San Nicolás de Hidalgo

• Thesis modality: Tesina Approved on October 3, 2025

Q COMMITES

2025

Master's thesis reviewer of Jaime Salvador López Viveros

• Thesis title: "Identificación de recombinación en virus utilizando análisis topológicos de datos"

• Posgrado Conjunto en Ciencias Matemáticas de la Universidad Nacional Autónoma de México y la Universidad Michoacana de San Nicolás de Hidalgo

2025

Master's thesis reviewer of Mariel Guadalupe Gutierrez Chaveste

• Thesis title: "Presentación de ADN mediante el juego de caos clasificación de imágenes y desarrollo de App de divulgación"

• Posgrado Conjunto en Ciencias Matemáticas de la Universidad Nacional Autónoma de México y la Universidad Michoacana de San Nicolás de Hidalgo

2025

Statistics General Knowledge Basic Exam

Centro de Ciencias Matemáticas, UNAM

• Semester 2025-2

Michoacan, Mexico

2024

Master's thesis reviewer of Juanita Claribel Santiago

• Thesis title: "Un teorema de caracterización para grupos hiperbólicos"

• Posgrado Conjunto en Ciencias Matemáticas de la Universidad Nacional Autónoma de México y la Universidad Michoacana de San Nicolás de Hidalgo

2024

Statistics General Knowledge Basic Exam

Centro de Ciencias Matemáticas, UNAM

• Semester 2025-1

Michoacan, Mexico

2024

Algebraic Topology General Knowledge Basic Exam

Centro de Ciencias Matemáticas, UNAM

• Semester 2024-2

Michoacan, Mexico

- 2023 • **Algebraic Topology General Knowledge Basic Exam**
Centro de Ciencias Matemáticas, UNAM  Michoacan, Mexico
• Semester 2024-1
- 2023 • **Algebraic Topology General Knowledge Basic Exam**
Centro de Ciencias Matemáticas, UNAM  Michoacan, Mexico
• Semester 2023-2



RESEARCH EXPERIENCE

- 2024 • **IAS Summer Collaborators Program**
IAS  Princeton, New Jersey
• Projects:
 - Lengths of the Orthogonal Geodesic Chords on Riemannian Manifolds.
 - Einstein Metrics on Homogeneous Manifolds.
 - Linear bounds for the lengths of geodesics on manifolds with curvature bounded below.
 • Catherine Searle, Megan Kerr, Regina Rotman, Erin Griffin, Isabel Beach
• July 2024
- 2023 • **Women in Geometry 3**
BIRS in Banff  Alberta, Canada
• Compact Homogeneous Einstein Manifolds
• Collaborators: Catherine Searle, Megan Kerr, Regina Rotman, Erin Griffin, Isabel Beach
• September 24 - 29
- 2023 • **Women in Computational Topology**
Bernoulli Center EPFL  Lausanne, Switzerland
• Studying self-similarity of complex networks with persistent magnitude
• Collaborators: Nina Otter, Maria Antonietta Pascali, Sanjukta Krishnagopal, Rayna Andreeva, Elizabeth Thompson
• July 17 - July 21
- 2022 • **2022 Summer Research in Mathematics program**
BIRS in Banff  Alberta, Canada
• Cohomogeneity Two Manifolds of Positive Sectional Curvature
• Collaborators: Catherine Searle, Megan Kerr, Regina Rotman, Hannah Alpert, Isabel Beach
• June 12 to June 25
- 2019 • **Scholarship Program for International Mobility for National CONACYT Fellows**
Department of Mathematics, The Ohio State University  Ohio, USA
• Dr. Jean-François Lafont
• December 1-15

- 2019 ● **Women in Geometry 2**
 Casa Matemática Oaxaca BIRS 📍 Oaxaca, Mexico
 • Cohomogeneity Two Manifolds of Positive Sectional Curvature
 • Collaborators: Catherine Searle, Megan Kerr, Regina Rotman, Hannah Alpert, Isabel Beach
 • June 23 to June 28
- 2018 ● **Scholarship Program for International Mobility for National CONACYT Fellows**
 Department of Mathematics, The Ohio State University 📍 Ohio, USA
 • Dr. Jean-François Lafont
 • April 30 to June 15

🗣️ TALKS AND POSTERS

- 2025 ● **XVI Taller de Álgebra y Topología y 3° Jornada Estudiantil de Ciencias Físico-Matemáticas**
 Invited talk 📍 Cuernavaca, México
 • De datos biológicos a invariantes topológicos
 • November 7th
- 2025 ● **Coloquio IMATE CU**
 Invited talk 📍 CDMX, Mexico
 • Explorando Datos Biológicos con Homología Persistente
 • October 28th
- 2025 ● **Prospectos en Topología**
 Talk 📍 Morelia, Mexico
 • Entropía e Información mutua
 • October 27th
- 2025 ● **1st Brazil-Mexico Joint Mathematical Meeting**
 Invited talk 📍 Fortaleza, Brazil
 • Quasi-isometric rigidity on 4-manifolds
 • Sesion on Geometry, Topology and Geometric Group theory
 • September 8th - 12
- 2025 ● **4° Congreso Latinoamericano de Ecología Microbiana, ISME Lat 2025**
 Talk 📍 Merida, Mexico
 • Exploring honey microbiomes: a metagenomic dashboard for taxonomy, biosynthesis, and antibiotic resistance
 • August 4th - 9th
- 2025 ● **Taller Mexicano de Topología**
 Invited talk 📍 CDMX, Mexico
 • Más allá de las formas: Cómo la topología revela patrones en datos biológicos
 • Session on Algebraic and Low-Dimensional Topology
 • June 17th

2025	● Mixtures of probability and Geometry	
	Keynote talk • Detecting Evolutionary Patterns and Cellular Organization through Persistent Homology • May 29th	📍 Oaxaca, México
2025	● Coloquio del Centro de Ciencias Matemáticas	
	Talk • Dimensiones fractales de redes complejas a través de TDA • May 9th	📍 Morelia, Mexico
2025	● Coloquio de Topología, Geometría y Aplicaciones	
	Minicourse • Análisis Topológico de Datos y Aplicaciones • April 28th - May 2nd	📍 Xalapa, México
2025	● Geometric Intelligence Workshop	
	Talk • How Topology Reveals Patterns in Biological Data • March 31th	📍 CDMX, Mexico
2025	● 2do Encuentro de las Matemáticas	
	Plenary talk • Más allá de las formas: Cómo la topología revela patrones en datos biológicos. • https://encuentromatematici1.wixsite.com/u-ecci-2025 • March 14th	📍 Bogotá, Colombia (Virtual)
2025	● La Matemática Atenagorica	
	Talk • Diversidad microbiana en mieles de abejas nativas sin aguijón de la Península de Yucatán • Co-authors: Xolalpa-Aroche, A., Hernández-Mena, D. I., Moguel-Chin, W. I., Rivero-Cruz, B. E., Ortiz-Vázquez, E., & Delgado-Suárez, E. J., Sélem-Mojica, N. • February 19th	📍 Michoacan, Mexico
2024	● 1st Asia-Pacific Bioinformatics Joint Conference	
	Poster • Microbial Diversity and Characterization of Stingless Bee Honey from the Yucatán Peninsula • Co-authors: Xolalpa-Aroche, A., Hernández-Mena, D. I., Moguel-Chin, W. I., Rivero-Cruz, B. E., Ortiz-Vázquez, E., & Delgado-Suárez, E. J., Sélem-Mojica, N. • October 26th	📍 Okinawa, Japan
2024	● XIX Seminario de Matemáticas Aplicadas UATx	
	Plenary talk • Análisis Topológico de Datos aplicado a la transferencia horizontal de genes de resistencia antimicrobiana • Co-authors: Shaday Guerrero-Flores, Nelly Sélem-Mojica. UATx Apizaco, Tlaxcala • September 23th	📍 Tlaxcala, Mexico

2024	● XIII Escuela de Verano en Matemáticas Talk • Magnitudes y Fractales • August 2nd	 Michoacan, Mexico
2024	● Escuela de Matemáticas de América Latina y el Caribe Cuernavaca 2024 Talk • Analizando datos con topología y geometría • Co-authors: Shaday Guerrero-Flores, Nelly Sélem-Mojica. IMATE Cuernavaca • June 18th	 Morelos, Mexico
2024	● Degustaciones Matemáticas Talk • TDA detecta la transferencia horizontal de genes • Co-authors: Shaday Guerrero-Flores, Nelly Sélem-Mojica. UMSNH • February 14 - 16	 Michoacan, Mexico
2024	● 2024 Joint Mathematics Meetings Talk • Asymptotic dimension and geometric decompositions in dimension 4 • Co-authors: Pablo Suárez-Serrato • January 6th	 San Francisco, USA
2023	● XXIV Escuela de Otoño (EOBM) y el XVII Encuentro Nacional (ENBM) de Biomatemáticas Minicourse • Aplicaciones de análisis topológico de datos a Pangenómica • October 9 - 14	 Veracruz, Mexico
2023	● Taller Propedéutico de Ingreso al PCCM 2023 Talk • Algunas aplicaciones relacionadas al análisis topológico de datos y aprendizaje geométrico profundo • October 9th	 Michoacan, Mexico
2023	● Iberoamerican and Pan Pacific International Conference on Topology and its Applications Talk • Invariants under quasi-isometries • September 11 - 14	 Puebla, Mexico
2023	● 31st Conference on Intelligent Systems for Molecular Biology and 22th European Conference On Computational Biology as part of the Annual International Conference on Critical Assessment of Massive Data Analysis Poster • Topology, regression and metapangenome enrich conventional microbiome analyses • July 24 - 27	 Lyon, France

2023	● Workshop Técnicas Modernas en Geometría Riemanniana 2023	
	Talk	📍 Mexico City, Mexico
	• Algunos invariantes de cuasi-isometrías • IMATE, UNAM • July 24 - 27	
2023	● Décima Jornada de Geometría, Topología y Dinámica	
	Talk	📍 Michoacan, Mexico
	• Algunos problemas relacionados al análisis topológico de datos y aprendizaje geométrico • CCM, UNAM • June 16th	
2023	● 2023 Midwest Geometry Conference	
	Talk	📍 Kansas, USA
	• Higher graph manifolds, properties and invariants • March 3 - 5	
2023	● 2023 Joint Mathematics Meetings	
	Talk	📍 Boston, USA
	• Collapsing and group growth as an obstruction to Einstein metrics on some smooth 4-manifolds • January 6th	
2022	● Seminario FISMAT	
	Talk	📍 Michoacan, Mexico
	• Algunos problemas relacionados al análisis topológico de datos y aprendizaje geométrico • Facultad de Ciencias Físico Matemáticas de la Universidad Michoacana de San Nicolás de Hidalgo • November 15th	
2022	● 8th Mexican Workshop on Applied Geometry and Topology	
	Poster	📍 Virtual
	• Topological data analysis applied to metagenomics and pangenomics data of <i>Clavibacter</i> • Co-authors: Shaday Guerrero Flores (Centro de Ciencias Matemáticas-CIMAT), Nelly Selem Mojica (Centro de Ciencias Matemáticas), Noé Bárcenas Torres (Centro de Ciencias Matemáticas), Jesús Hernández Hernández (Centro de Ciencias matemáticas), Antonio Rieser (CIMAT) • November 11th	
2022	● 8th Mexican Workshop on Applied Geometry and Topology	
	Poster	📍 Virtual
	• Topological and Geometric explorations of Autism fMRI connectomes • Co-authors: Sebastian Pujalte Ojeda (Instituto de Fisiología Celular UNAM), Sophie Achard (Centre de recherche Inria Grenoble Rhone-Alpes), Pablo Suárez Serrato (Instituto de Matemáticas UNAM) • November 10th	

2022	● ISCB-Latin America SoIBio BioNetMX Conference on Bioinformatics 2022	📍 Queretaro, Mexico
	Poster • Topological data analysis applied to metagenomics and pangenomics data of <i>Clavibacter</i> • Co-authors: Shaday Guerrero Flores (Centro de Ciencias Matemáticas-CIMAT), Nelly Selem Mojica (Centro de Ciencias Matemáticas), Noé Bárcenas Torres (Centro de Ciencias Matemáticas), Jesús Hernández Hernández (Centro de Ciencias matemáticas), Antonio Rieser (CIMAT) • November 3rd	
2022	● 55 Congreso Nacional de la Sociedad Matemática Mexicana	📍 Guadalajara, Mexico
	Talk • Cuasi-isometrías e invariantes • October 26th	
2022	● Coloquio del Posgrado Conjunto de Ciencias Matemáticas	📍 Michoacan, Mexico
	Talk • Variedades de Gráficas y Métricas de Einstein • September 23th	
2022	● Seminario Informal de Geometría del CCM	📍 Michoacan, Mexico
	Talk • Cuasi-isometrías y “aplicaciones” • September 22th	
2022	● XXII Escuela de Verano en Matemáticas	📍 Michoacan, Mexico
	Talk • Cuasi-Isometrías y algunos invariantes • CCM, UNAM • August 3rd	
2022	● I Escuela de Teoría Geométrica de Grupos	📍 Michoacan, Mexico
	Talk • ¿Qué significa crecimiento? • CCM, UNAM • July 28th	
2022	● Seminario DIVAGEO	📍 Virtual
	Talk • Variedades de gráficas, propiedades y métricas de Einstein • Facultad de Ciencias, UNAM • May 13th	
2022	● Seminario a distancia de Teoría Geométrica de Grupos	📍 Virtual
	Talk • Variedades de Gráficas de dimensiones superiores, propiedades e invariantes • CCM, UNAM • May 5h	

2021	● Tercer Encuentro de Mujeres Matemáticas Mexicanas Talk • Sobre Variedades de Gráficas • November 16 - 19	📍 Virtual
2021	● 54 Congreso Nacional de la Sociedad Matemática Mexicana Talk • Sobre Variedades de Gráficas • October 18 - 22	📍 Virtual
2021	● Geometric Group Theory seminar in Cambridge Talk • About higher graph manifolds • May 21th	📍 Virtual
2021	● Spring Eastern Sectional Meeting, Recent Developments in Differential Geometry, III, Talk • Quasi-isometric rigidity in cusp-decomposable manifolds • https://sites.google.com/wellesley.edu/amsspecialsession2021/home • March 20 - 21	📍 Virtual
2021	● Sextas Jornadas de Geometría, Topología y Dinámica Talk • Sobre las variedades de gráficas de dimensiones superiores • March 25th	📍 Virtual
2020	● Workshop and Winter School on Geometric and Topological Data Analysis Poster • Topological gastronomy • CIMAT • January 23th	📍 Guanajuato, Mexico
2019	● Seminar ATD and related topics in CIMAT Talk • Topological gastronomy • CIMAT • November 1st	📍 Guanajuato, Mexico
2019	● LII Congreso Nacional de la Sociedad Matemática Mexicana Talk • Rigidez Casi-Isométrica en Variedades de Gráficas Superiores • October 24th	📍 Nuevo Leon, Monterrey

- 2019 ● **Workshop Geometría Diferencial y Rigidez de espacios singulares**
Talk
• Rigidez Casi-Isométrica en Variedades de Gráficas Superiores
• CCM, UNAM
• June 16th
- 📍 Michoacan, Mexico



AWARDS

- 2030
|
2026 ● **Investigador Nacional Nivel I**
Sistema Nacional de Investigadoras e Investigadores
- 2029
|
2025 ● **CONAHCYT Postdoctoral Research Fellowship**
- 2023
|
2022 ● **CONACYT Ciencia Frontera Postdoctoral Research Fellowship**
- 2023 ● **1st place at the Annual International Conference on Critical Assessment of Massive Data Analysis**
ISCB
• Anti-Microbial Resistance Prediction and Forensics in urban metagenomes
• July 2023
📍 Lyon, France
- 2022 ● **Mentee in the Mentoring in Action – Women and Girls in STEM Online programme**
British Council, Inova Consultancy, WiTEC
• Humiko Yahaira Hernández Acosta
• April - June 2022
📍 Mexico
- 2022 ● **2nd place at the ISCB-La SOLBIO BIONETMX WIKIPEDIA COMPETITION**
ISCB
• Matrices de distancias
• October, 2022
📍 Queretaro, Mexico
- 2020
|
2016 ● **CONACYT PhD Fellowship**
- 2018 ● **CONACYT Research Fellowship**
Ohio State University
• Scholarship Program for International Mobility for National CONACYT Fellows
📍 Ohio, USA
- 2016
|
2014 ● **CONACYT Masters Fellowship**
- 2011 ● **Academia Mexicana de Ciencias Summer Research Fellowship for undergraduate students**
Universidad Michoacana de San Nicolas Hidalgo
• Dr. Jorge Luis López López
• 3-variedades hiperbólicas
📍 Michoacan, Mexico

- 2010 ● **Academia Mexicana de Ciencias Summer Research Fellowship for undergraduate students**
Universidad Michoacana de San Nicolas Hidalgo  Michoacan, Mexico
- Dr. Jorge Luis López López
 - 3-variedades hiperbólicas

SCIENCE COMMUNICATION

- 2025 ● **Feria Synapsia**
Workshop  Morelia, Mexico
- Taller de Papiroflexia Modular
 - Coordinator of the Workshop for middle and high school students
 - October 16th - 17th
- 2025 ● **Segundo Encuentro de Mujeres en Topología, Geometría y sus Aplicaciones**
Workshop  Guanajuato, Mexico
- Taller de Papiroflexia y Origami
 - Coordinator of the Workshop for middle and high school students as part of the Second Meeting
 - October 3th
- 2025 ● **XXXIII Tianguis de la Ciencia**
Workshop  Morelia, Mexico
- Taller de Papiroflexia Modular
 - Coordinator of the Workshop
 - June 6th - 7th
- 2025 ● **Feria de las Matemáticas**
Workshop  Michoacan, Mexico
- Taller de Papiroflexia Modular
 - Coordinator of the Workshop
 - March 15th
- 2025
|
2023 ● **Club Mate**
Workshop  Michoacan, Mexico
- Participation in the mate club aimed at elementary and secondary school students
 - CCM UNAM
- 2024 ● **Futuras STEM Michoacán**
Workshop  Michoacan, Mexico
- Taller de Matematica
 - December 5th
- 2024 ● **12° Fiesta de las Ciencias y Humanidades**
Workshop  Michoacan, Mexico
- Taller de Matematica
 - November 14th-15th

2024	● Feria Infantil y Juvenil de Ciencias y Artes Workshop • Estereogramas • May 25th	 Michoacan, Mexico
2024	● Feria de las Matemáticas Workshop • Taller de Papiroflexia Modular • Coordinator of the Workshop • March 16th	 Michoacan, Mexico
2024	● Feria de las Matemáticas Talk • ¿Qué hacemos cuentas? Son puros cuentos • Co-authors: Cristina Villanueva, Yesenia Villicaña, Diana Piñango • March 16th	 Michoacan, Mexico
2023	● PI Day in the Instituto Tecnológico Superior de Martínez de la Torre Talk • Algunas aplicaciones relacionadas al análisis topológico de datos y aprendizaje geométrico • ITSMT, Mtz. de la Torre. • March 14th	 Veracruz, Mexico
2023	● 11° Fiesta de las Ciencias y Humanidades Workshop • Lengua de Señas Mexicana (LSM) y Matemáticas • UNAM Campus Morelia • October 19th	 Michoacan, Mexico
2023	● 11° Fiesta de las Ciencias y Humanidades Workshop • Biomatemáticas • UNAM Campus Morelia • October 20th	 Michoacan, Mexico
2022	● 10° Fiesta de las Ciencias y Humanidades Workshop • Lengua de Señas Mexicana (LSM) y Matemáticas • UNAM Campus Morelia • October 27th - 28th	 Michoacan, Mexico



SYNERGISTIC ACTIVITIES

current
|
2024



Cryptography Seminar

📍 Michoacan, Mexico

- Organizer of the Cryptography Seminar at the CCM Morelia

current
|
2022



Long-distance Seminar on Geometric Group Theory in Mexico

📍 Virtual

- Co-organizer of Long-distance Seminar on Geometric Group Theory in Mexico

current
|
2022



R Ladies Morelia

📍 Michoacán, Mexico and hybrid

- Co-organizer of the sessions of R Ladies Morelia <https://r-ladies-morelia.github.io/> https://www.meetup.com/es-ES/r-ladies-morelia/?eventOrigin=home_groups_you_organize
- CCM UNAM

2025



Segundo Encuentro de Mujeres en Topología, Geometría y sus Aplicaciones

📍 Guanajuato, Mexico

- Co-organizer with Sandy Aguilar, Yesenia Villicaña Molina, Teresa Hoekstra, Raquel Perales and Ferrán Valdez Webpage: <https://matmor.unam.mx/encuentro-mujeres-topogeo-2025/>
- CCM UNAM - CIMAT - CWM
- September 30th - October 3rd

2025



IV Escuela de Teoría Geométrica de Grupos

📍 Michoacan, Mexico

- Co-organizer of IV Escuela de Teoría Geométrica de Grupos for undergraduate and graduate students. Web page: <https://matmor.unam.mx/etgg-2025/>
- CCM UNAM
- August 4th - 8th

2024



Primer Encuentro de Mujeres en Topología, Geometría y sus Aplicaciones

📍 Michoacán, Mexico

- Co-organizer with Sandy Aguilar, Yesenia Villicaña Molina, Ferrán Valdez Webpage: <https://www.matmor.unam.mx/encuentro-mujeres-topogeo/>
- CCM UNAM
- September 11th - 13th

2024



III Escuela de Teoría Geométrica de Grupos

📍 Michoacan, Mexico

- Co-organizer of III Escuela de Teoría Geométrica de Grupos for undergraduate and graduate students
- CCM UNAM
- July 22th - 26th

2023



Encuentro Nacional de R ladies México

📍 Virtual

- Co-organizer of Encuentro Nacional de R ladies México
- September 23

- 2023 ● **II Escuela de Teoría Geométrica de Grupos**  Michoacan, Mexico
- Co-organizer of II Escuela de Teoría Geométrica de Grupos for undergraduate and graduate students
 - CCM UNAM
 - June 19 th - 23th
- 2023 ● **Hackaton Reto Critical Assessment of Massive Data (CAMDA)**  Michoacan, Mexico
- Co-organizer of Hackaton Reto Critical Assessment of Massive Data (CAMDA) de geolocalización de datos de microbioma
 - CCM UNAM
 - June 3rd - 7th
- 2022 ● **Hackaton TDA**  Michoacan, Mexico
- Co-organizer of the hackaton Análisis Topológico de Datos en el Hackaton Aplicaciones de Matemáticas en problemas de datos biológicos
 - CCM UNAM
 - November
- 2021 ● **Seminario de Ciencias Básicas**  Veracruz, Mexico
- Organizer and Co-organizer of the Seminario de Ciencias Básicas in the Instituto Tecnológico Superior de Martínez de la Torre, Veracruz for undergraduate students



WORKSHOPS AND COURSES ATTENDED

- 2023 ● **1st Advanced Scientific Programming in Python – Latin America**  Mexico City, Mexico
- Main instructors: Carlos Cernuda, Carlos Echeverría, Guillermo Aguilar, Jakob Jordan, Jenni Rinker, Lisa Schwetlick, Pietro Berkes, Rike-Benjamin Schuppner, Seetha Krishnan, Tiziano Zito, Zbigniew Jędrzejewski-Szmek
 - <https://latam.aspp.school/2023-mexico-city/start.html>
 - June 26th - July 1st
- 2023 ● **Taller Descarga y Análisis de Datos Digitales (Facebook)**  Virtual
- October 25th



CERTIFICATIONS

- 2024 ● **The Carpentries Instructor Training Program**  Online
- The Software Carpentry Foundation

July 2022	EC0553 Comunicación efectiva en el trabajo Consejo de Normalización y Certificación de Competencia Laboral (CONOCER) Martínez de la Torre, Mexico
July 2022	EC0076 Evaluación de la competencia de candidatos con base en Estándares de Competencia Consejo de Normalización y Certificación de Competencia Laboral (CONOCER) Martínez de la Torre, Mexico
November 2021	Inducción a la Normatividad que regula la relación laboral con el TecNM Consejo de Normalización y Certificación de Competencia Laboral (CONOCER) Martínez de la Torre, Mexico • CRODE Celaya
October 2021	EC0772 Evaluación del aprendizaje con enfoque en competencias profesionales Consejo de Normalización y Certificación de Competencia Laboral (CONOCER) Martínez de la Torre, Mexico
2021	Sistemas de Gestión de Calidad ISO 9001:2015 Instituto Tecnológico de Martínez de la Torre, TecNM Martínez de la Torre, Mexico • Capacitación en fundamentos de gestión de calidad bajo ISO 9001:2015 aplicada a instituciones educativas. • Participación en procesos de auditoría interna, en la implementación y documentación de procesos administrativos-académicos.



TRAINING AND WORK RELATED TO EQUALITY

2021	Masculinidades respetuosas de los derechos humanos Instituto Veracruzano de las Mujeres Online
2021	Jornada de capacitación para el regreso seguro a clases. Secretaría de Educación de Veracruz Online
2021	Educación superior: retorno seguro. IMSS Online
2021	Acciones y rutas para prevenir y atender la violencia contra las mujeres Instituto Veracruzano de las Mujeres Online



ADDITIONAL PROFESSIONAL TRAINING (FIELDWORK SAFETY AND FIRST AID)

2023	Manejo y uso de botiquín en prácticas de campo ENES Morelia, UNAM Morelia, Mexico
2022	Prevención y combate de fuego ENES Morelia, UNAM Morelia, Mexico

2022



Primeros auxilios básicos
ENES Morelia, UNAM

Morelia, Mexico